Amey Mishra

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Education and Training

HARMONY SCHOOL OF ADVANCEMENT | Houston, TX | Expected May 2027

High School Diploma | GPA: 4.74 | Valedictorian | SAT: 1520 | ACT: 35

- Senior: Multivariable Calculus, Linear Algebra, AP Research, AP Cybersecurity, AP Environmental Science, AP English Literature.
- Completed: AP Calculus BC, AP Statistics, AP Biology, AP Computer Science A, AP U.S. History, AP Seminar, AP English Language.

Technical Skills

Mechanical Design / Fabrication — Fusion 360, Onshape, Autodesk Revit, mechanism design, CAD assemblies, 3D printing, prototyping, actuator integration, design for manufacture.
Programming / Controls — Python, Java, Arduino, JavaScript, OpenCV, sensor feedback, autonomous navigation, path planning, inverse kinematics, control-system testing.
Engineering / Research — Experimental design, simulation, data analysis, orbital mechanics, literature review, feasibility assessment, technical presentation.

Experience

FOUNDER, CAPTAIN & MECHANICAL LEAD | 06/2023 - Present | 15 hrs/wk, 45 wks/yr

FIRST Tech Challenge Team 25679 The Cosmobots | 25943 HIVE Robotics — Houston, TX | thecosmobots.com

- Led design, prototyping, fabrication, testing, strategy, and team operations.
- Among the top teams of 8,000+ worldwide; 4th-ranked alliance at Worlds; Carolinas Premier Event winner; FTC Hall of Fame; FIRST Leadership Award Finalist (peer-nominated, internationally judged).
- Redesigned high-cycle scoring mechanisms via iterative CAD, fabrication, and field testing; cut transfer time from ~0.75 s to 0.30 s and improved consistency.
- Used surgical tubing, constrained geometry, actuator-placement studies, and serviceability reviews to simplify mechanisms and boost reliability.
- Built autonomy/controls with custom path planning, feedback loops, and mechanism libraries.
- Ran design reviews, timelines, scouting/strategy, competition prep, sponsors, and mentors.

FOUNDER & CEO | 06/2024 - Present | 10 hrs/wk, 50 wks/yr

Cosmobots Robotics — Houston, TX | cosmobotsrobotics.com

- Founded and scaled a STEM nonprofit for robotics, rocketry, mentorship, and outreach; launched or supported 4 FTC and 3 American Rocketry Challenge teams.
- Secured $50,000+ in sponsorships, grants, and in-kind support; managed budgets, equipment, travel, registrations, mentors, volunteers, and long-term planning.
- Partnered with Unitree Robotics, Stokes Education, Harmony Public Schools, Meiborg, and others to expand equipment access and programming.
- Logged 4,500+ service hours; ran 30+ demos/workshops for Title I and underserved communities.
- Reached 25,000+ people and students in 7 countries through demos, resources, and mentorship.

Research and Professional Experience

LEAD RESEARCHER & SYSTEMS ENGINEER | 2025 - Present | 5 hrs/wk, 26 wks/yr

Space Debris Research / NASA Space Apps Challenge — ORCA | orbitaldebriscleanup.space

- Published a study in the American Journal of Student Research on orbital-debris growth, collision risk, impacts, and mitigation/removal.
- Led systems engineering for ORCA (Orbital Recycling and Construction Array), which captures debris and repurposes usable material instead of disposing of it.
- Trained a neural network for debris analysis/pathing; evaluated dataset candidates.
- Modeled capture paths and mission geometry with orbital mechanics; CAD-designed and simulated debris collection, handling, and recycling mechanisms.
- Built feasibility, implementation, operations, and business-plan components.
- NASA Space Apps Regional Winner and Global Semifinalist; accepted to IEEE ISMSIT 2026 as corresponding author/speaker (IEEE Xplore pending).

INDEPENDENT ROBOTICS & AI RESEARCHER | 2026 - Present | 6 hrs/wk, 12 wks/yr

Mentored by a Stanford PhD Student

- Built a neural-network-powered humanoid arm: 3D-printed parts, motors, sensors, embedded control.
- CAD-iterated for actuator placement, joint range, structure, and manufacturability.
- Adaptive inverse kinematics and Java/Arduino controls turn targets into joint commands.
- Integrated OpenCV vision to use environmental data during movement and interaction.
- Refined hardware, sensors, controls, and model via testing, calibration, and redesign.

DESIGN & MANUFACTURING INTERN | 07/2026 - Present | 8 hrs/wk, 12 wks/yr

Pike Robotics

- Built motion systems for a robot that inspects industrial tanks 100+ ft in diameter.
- Designed pivoting camera mounts and umbilical management in Fusion 360 and Onshape.
- Compared concepts for motion range, packaging, serviceability, reliability, and manufacturability; refined them via CAD iteration and design review under real constraints.

SUMMER RESEARCH INTERN | Summer 2026 | 5 hrs/wk, 6 wks/yr

Rice University — Electromagnetic Energy Harvesting Research

- Derived tuning-state laws for electromagnetic vibration harvesters; evaluated them with 8,400 simulated speed profiles and machine data.
- Programmed microcontroller test rigs; simulated prototypes under changing conditions.
- Analyzed feasibility/performance tradeoffs; produced a presentation and manuscript.
- Presented at the London International Conference; submitted to the Journal of Micromechanics and Microengineering.

Honors and Awards

FIRST Leadership Award Finalist — Peer-nominated, internationally judged.
FIRST Tech Challenge — 4th-ranked World Championship alliance; Hall of Fame; Carolinas Premier Champion.
NASA Space Apps — Regional Winner and Global Semifinalist for ORCA.
RECF Aerial Drone Nationals — Flight Operations & Judges Awards; 3x Nationals qualifier.
Gene Kranz Scholarship — NASA Space Center U; Thermodynamics winner, Cryogenics runner-up.
International Math Challenge — Gold Award (previously Silver).
Science Olympiad — 1x State Gold, 2x State Bronze, 7x Regional Gold (engineering/build).

Engineering, Competition and Leadership

TEAM FOUNDER, CAPTAIN & PILOT | 08/2023 - Present | 4 hrs/wk, 32 wks/yr

RECF Aerial Drone Competition — Teams 96610A / 96610B

- Managed two teams: equipment procurement, registration, practice planning, FAA TRUST compliance, safety training, travel, and logistics.
- Developed a reusable Python autonomous-flight library using sensor feedback loops and navigation logic for repeatable obstacle-course and mission runs.
- Built autonomous flight programs with ~97% successful execution in testing, refined through repeated ground and controlled tests.
- Combined autonomous routines with manual flight strategy, refining control logic around course constraints and scoring objectives.
- Earned 4 Regional Championships, 3 Regional All-Around/Excellence Awards, 3 National Championship qualifications, and the national Flight Operations and Judges Awards.
- Mentored regional teams and students on flight control, programming, safety, and competition prep.

TEAM LEAD / FLIGHT SYSTEMS DEVELOPER | 2025 - Present | 2 hrs/wk, 20 wks/yr

American Rocketry Challenge | Independent Thrust-Vector-Control Development

- Led rocket design, construction, testing, launch prep, budgeting, travel, and team operations for competition and outreach launches.
- Designed, built, and tested a custom thrust-vector-control system using CAD, 3D printing, electronics, actuation, and flight-data analysis.
- Integrated FAA TRUST, safety training, launch procedures, and hardware checks; pursuing supervised National Association of Rocketry Level 1 certification.

PRESIDENT & ENGINEERING EVENT COMPETITOR | 2023 - Present | 4 hrs/wk, 30 wks/yr

Harmony School of Advancement Science Olympiad

- Led a 15-member team: assigned events, coordinated prep schedules, and managed build, testing, and competition logistics.
- Specialized in build events: Tower, Air Trajectory, Flight, Helicopter, Electric Vehicle, and Storm the Castle.
- Earned 7 Regional Gold medals in engineering/build events, plus 1 State Gold and 2 State Bronze.
- Applied structural design, aerodynamics, mechanics, electronics, iterative testing, and data-driven optimization.

STUDENT COUNCIL PRESIDENT | 2023 - 2025 | 2 hrs/wk, 30 wks/yr

Harmony School of Advancement Student Council

- Represented ~800 students; led seven schoolwide events, service initiatives, engagement programs, and Class of 2027 fundraising.
- Coordinated officers, faculty sponsors, logistics, communications, and execution with input from the student body.

COMPETITOR & INDEPENDENT SOLVER | 2023 - Present | 2 hrs/wk, 30 wks/yr

Competitive Speedcubing / World Cube Association

- Practiced advanced methods incl. COLL; official WCA personal bests of 9.38 s (3x3) and 2.07 s (2x2).
- Won the Harmony competition twice and was a finalist twice; taught younger students solving, pattern recognition, and problem-solving.

Engineering Projects | 6 hrs/wk, 52 wks/yr (combined)

AUTONOMOUS MAPPING UAV

Independent Engineering Project

- CAD-designed and 3D-printed UAV for waypoint navigation and location mapping.
- Integrated autonomous navigation, a high-res camera, and live video downlink.
- Iterated airframe, electronics, payload packaging, and controls via flight testing.
- Unified mechanical design, electronics, navigation software, and flight ops.

RC AERODYNAMICS TEST AIRCRAFT

Independent Aerospace Engineering Project

- Designed and 3D-printed an RC aircraft; tested flaps and spoilers for handling and stability.
- Linked aerodynamics to observed behavior through flight tests, repairs, and redesign.
- Used qualitative and flight-performance observations to guide design changes.

AUTONOMOUS LAWN MOWER

Independent Robotics Project

- Developed an autonomous lawn mower to cut repetitive manual work; integrated drive systems, electronics, sensing, and controls.
- Iterated chassis/layout for traction, packaging, safety, maintenance, wiring, and field use.

HUMANOID ROBOT

Independent Robotics Project

- Built a 3D-printed humanoid with Arduino motor control, OpenCV perception, API integration, facial recognition, kinematics-based movement, and task scheduling.
- Full-stack: mechanical design, embedded control, perception, software, human-like motion.
- Co-iterated hardware and code to improve responsiveness and motion accuracy.

NASA EXOPLANET ANALYSIS

Independent Machine Learning Project

- Trained neural nets (incl. a custom CNN) on public NASA data to classify exoplanet candidates.
- Compared training iterations; tuned the pipeline for accuracy and fewer inconsistent results.

ECOSCAPE SUSTAINABLE ARCHITECTURE

eCYBERMISSION Engineering Project

- Designed a sustainable city from current tech, environmental data, and ecological analysis.
- Adapted the design to varied climates; Texas State 2nd Place in eCYBERMISSION.

3D PRINTING SERVICES

Self-Employed

- Paid 3D-printing service: turned client needs into manufacturable CAD and prints.
- Managed slicing, materials, production, troubleshooting, and part refinement.
- Balanced accuracy, print time, strength, and material use in print settings.

Certifications

Engineering / IT — Autodesk Fusion 360 & Revit; CompTIA Network+; OpenEDG & Microsoft ITS Python.
Safety / Technical — FAA Recreational Pilot Certificate; Photoshop & Illustrator; W3Schools JavaScript.

Additional Leadership and Service

STEM Mentoring — Mentored robotics, Science Olympiad, drone, and engineering students in design reviews, troubleshooting, programming, and competition prep.
The Congressional Award — Congressional Service Award – Gold, highest honor for civilian youth.